

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			Alternative 2 – Substitution and rearrange i.e. $I = \frac{4.72 \times 10^{28} \text{ecf}}{4\pi \times (37 \times 9.46 \times 10^{15})^2} \text{ (1)}$ = 3.07 × 10⁻⁸ [W m⁻²] (1) Compare with 3.09 × 10⁻⁸ [W m⁻²] so claim correct or valid conclusion if ecf used Alternative 3 – Substitution and rearrange i.e. $P = 3.09 \times 10^{-8} \times 4\pi \times (37 \times 9.46 \times 10^{15})^2 \text{ (1)}$ = 4.72 × 10²⁸ [W] Compare intensities so claim correct or valid conclusion if ecf used (1) To summarise: Correct substitution into Stefan's law (1) Quoting or using $I = \frac{P}{4\pi d^2}$ (1) Correct substitution and algebra (1) Correct almost final value (1) Comparison and valid comment (1)						
			Question 7 total	4	3	5	12	9	0